

AI Recipe Generator

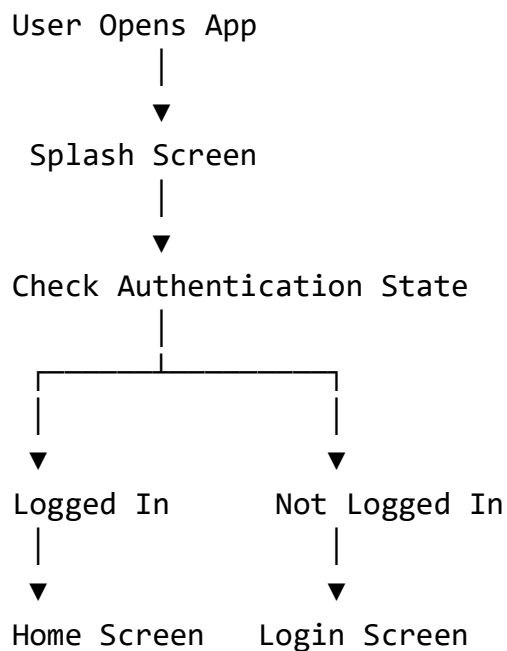
Application Flow Document (AFD)

Version: 1.0

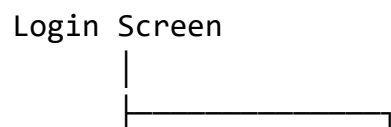
Platform: Flutter (Android)

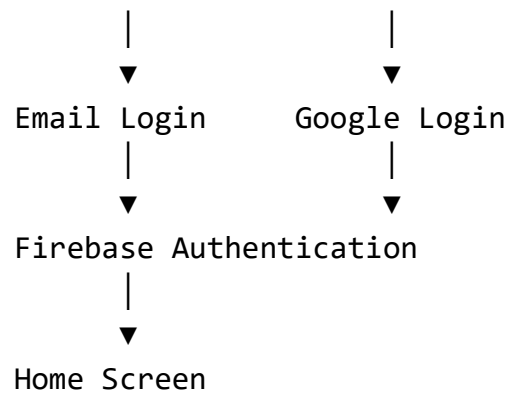
Architecture: Clean Architecture + Provider + Firebase + Gemini AI

1. Application Launch

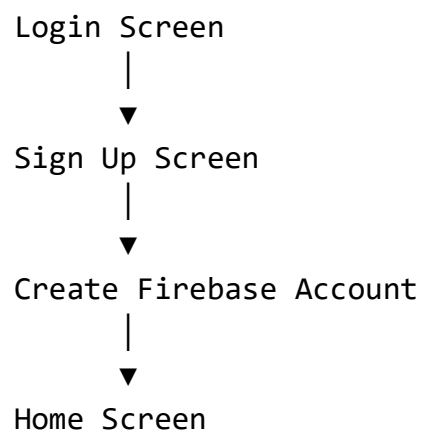


2. Authentication Flow

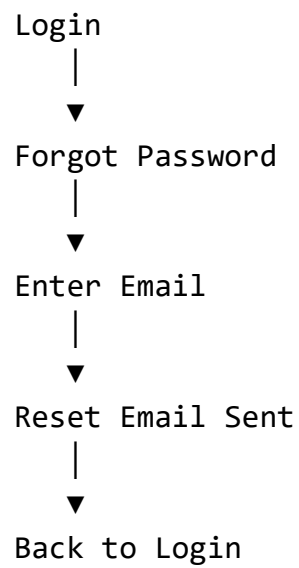




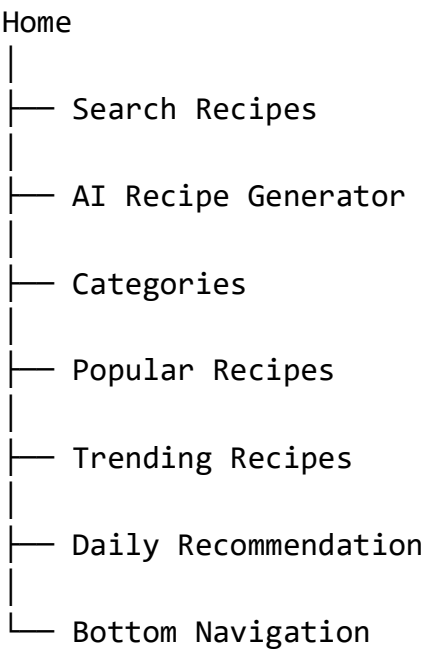
Alternative Navigation



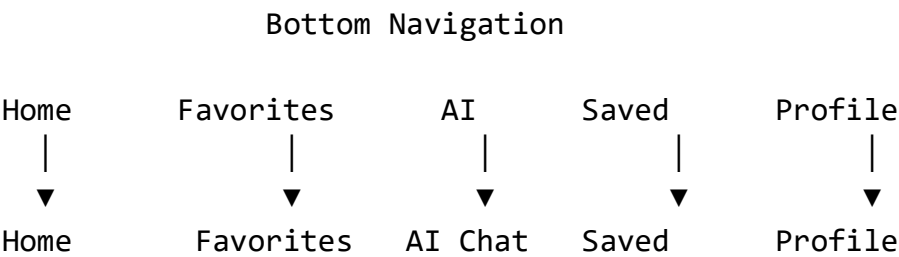
Forgot Password Flow



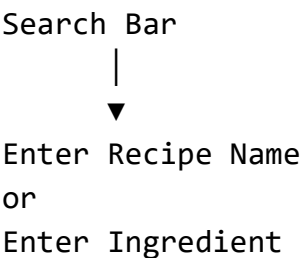
3. Home Screen Flow

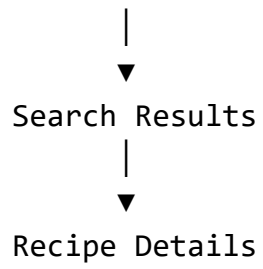


4. Bottom Navigation Flow



5. Search Flow





6. Category Flow

Categories

Breakfast

Lunch

Dinner

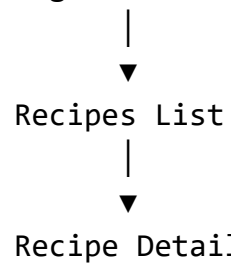
Desserts

Healthy

Snacks

Drinks

Vegan



7. AI Recipe Generator Flow

AI Generator



User enters prompt

Example:

"Make a healthy chicken dinner under 500 calories"



Gemini API



Generate Recipe



Recipe Details



Save Recipe

Share Recipe

8. AI Chat Flow

AI Chat



User asks cooking question



Gemini AI

|
▼
AI Response

|
▼
Continue Conversation

Examples

How do I cook steak?

What can I make with potatoes?

Healthy breakfast ideas

Vegetarian dinner recipes

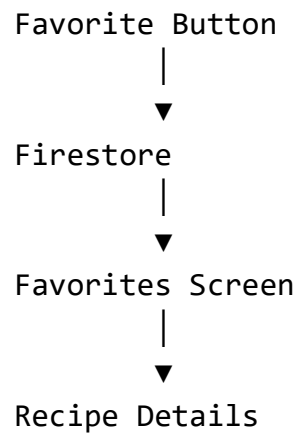
9. Recipe Details Flow

Recipe Details

|
├─ Recipe Image
├─ Title
├─ Cooking Time
├─ Difficulty
├─ Calories
├─ Ingredients
├─ Instructions
├─ Nutrition
├─ AI Tips
|
├─ Favorite
├─ Save
├─ Share

|
└─ Back

10. Favorites Flow



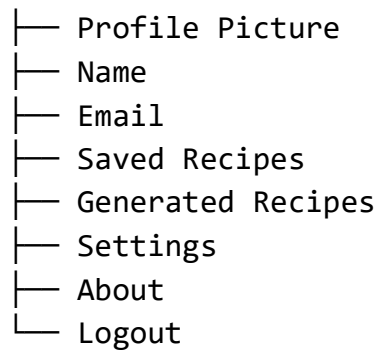
11. Saved Recipes Flow



12. Profile Flow

Profile

|



Logout

Logout



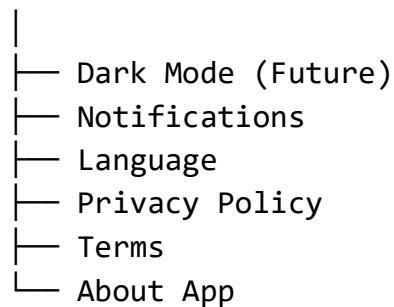
Firebase Sign Out



Login Screen

13. Settings Flow

Settings



14. Error Flow

No Internet



Show Error Screen



Retry Button



Reconnect

Authentication Error

Wrong Password

Email Not Found

Weak Password

Account Already Exists

Display Snackbar

AI Error

Gemini Timeout

API Limit

Invalid Response

Show Retry Button

15. Complete User Journey

Splash

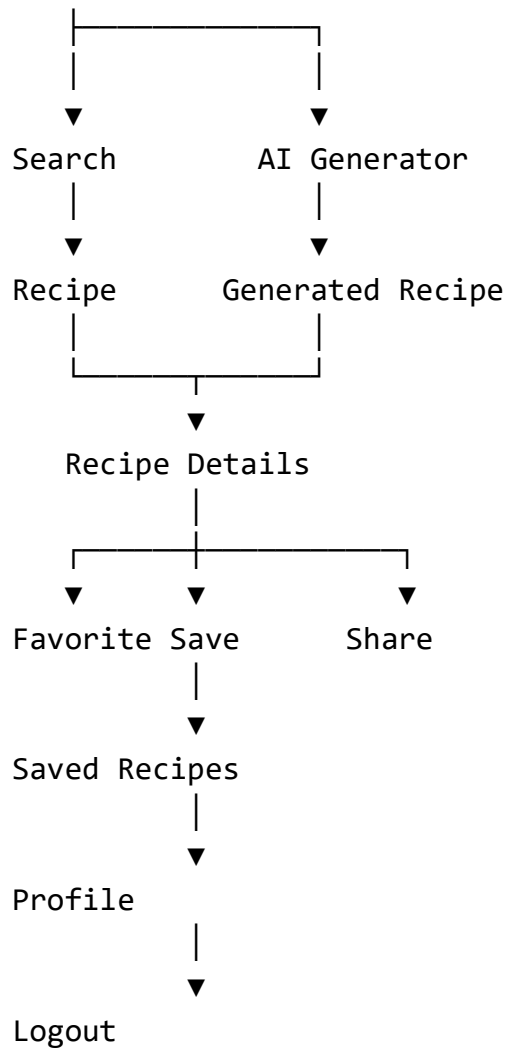


Login

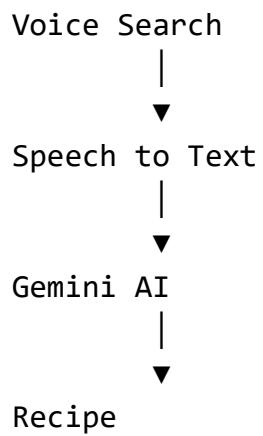


Home





16. Future Feature Flow (Phase 2)



Camera



▼
Ingredient Detection



▼
Gemini



▼
Suggested Recipes

Meal Planner



▼
Select Days



▼
AI Weekly Meal Plan

Shopping List



▼
Generate Ingredients



▼
Shopping Checklist